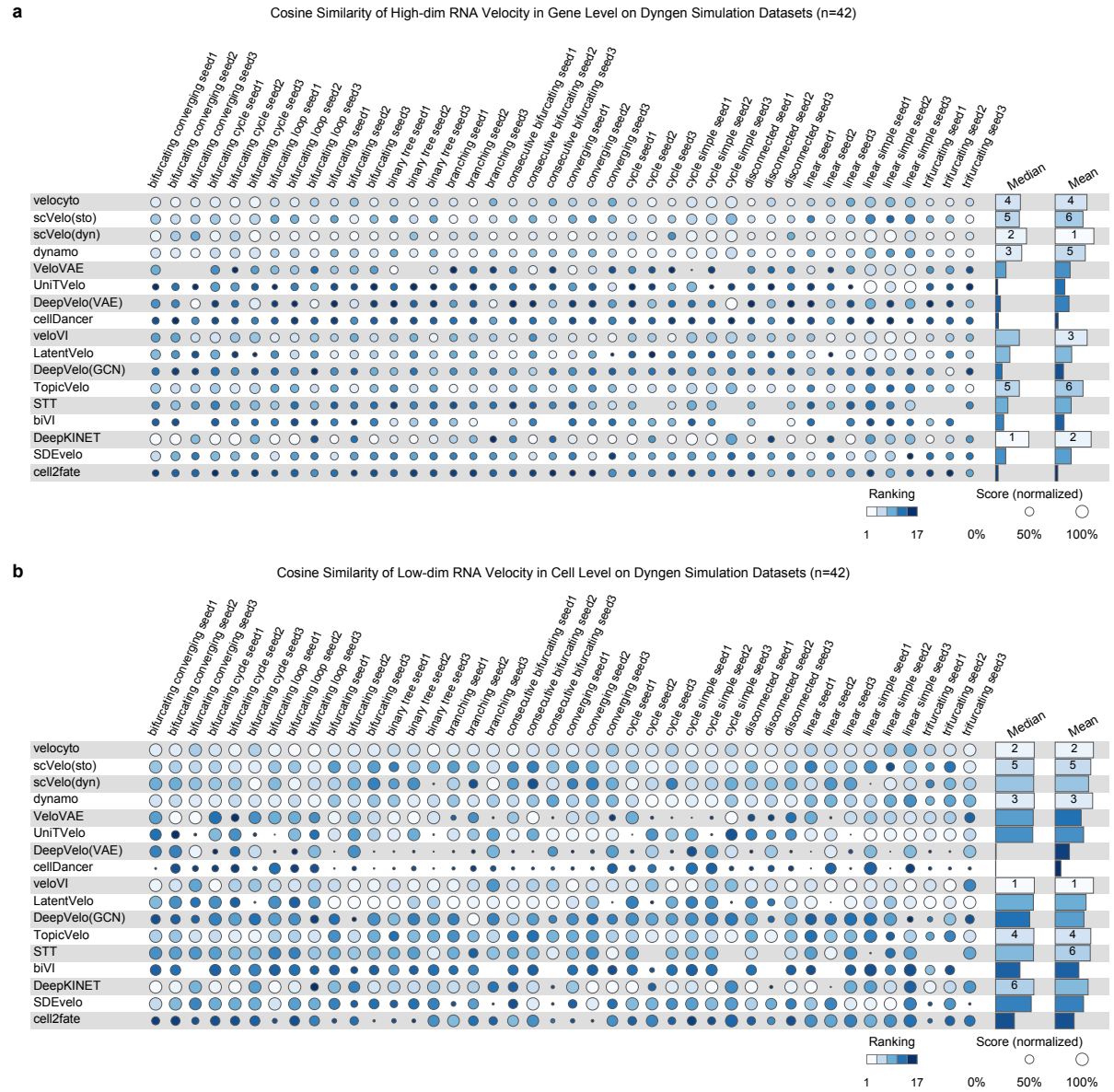

a
Cosine Similarity of High-dim RNA Velocity in Gene Level on Dyngen Simulation Datasets (n=42)
bifurcating converging seed1
bifurcating converging seed2
bifurcating converging seed3
bifurcating cycle seed1
bifurcating cycle seed2
bifurcating cycle seed3
bifurcating loop seed1
bifurcating loop seed2
bifurcating loop seed3
bifurcating seed1
bifurcating seed2
bifurcating seed3
binary tree seed1
binary tree seed2
binary tree seed3
branching seed1
branching seed2
branching seed3
consecutive bifurcating seed1
consecutive bifurcating seed2
consecutive bifurcating seed3
converging seed1
converging seed2
converging seed3
cycle seed1
cycle seed2
cycle seed3
cycle simple seed1
cycle simple seed2
cycle simple seed3
disconnected seed1
disconnected seed2
disconnected seed3
linear seed1
linear seed2
linear seed3
linear simple seed1
linear simple seed2
linear simple seed3
trifurcating seed1
trifurcating seed2
trifurcating seed3
Median
Mean
velocyto 4 4
scVelo(sto) 5 6
scVelo(dyn) 2 1
dynamo 3 5
VeloVAE
UniTVelo
DeepVelo(VAE) 5 6
cellDancer
veloVI 3
LatentVelo
DeepVelo(GCN)
TopicVelo
STT
biVI
DeepKINET 1 2
SDEvelo
cell2fate
Ranking
1
17
Score (normalized)
0%
50%
100%
b
Cosine Similarity of Low-dim RNA Velocity in Cell Level on Dyngen Simulation Datasets (n=42)
Median
Mean
velocyto 2 2
scVelo(sto) 5 5
scVelo(dyn)
dynamo 3 3
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI 1 1
LatentVelo
DeepVelo(GCN)
TopicVelo 4 4
STT 6
biVI
DeepKINET 6
SDEvelo
cell2fate
Ranking
1
17
Score (normalized)
0%
50%
100%
c
Mono-kinetic
Transcriptional boost
Multi-forward branching
Cell-Level Cosine Similarity
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
velocyto UniTVelo scVelo(dyn) SDEvelo STT cellDancer dynamo veloVI scVelo(sto) DeepVelo(GCN) LatentVelo biVI DeepVelo(VAE) VeloVAE
VeloVAE cellDancer dynamo velocyto UniTVelo SDEvelo DeepVelo(GCN) scVelo(dyn) scVelo(sto) veloVI biVI DeepVelo(VAE) LatentVelo STT
velocyto dynamo VeloVAE cellDancer scVelo(sto) SDEvelo DeepVelo(VAE) scVelo(dyn) DeepVelo(GCN) UniTVelo veloVI biVI LatentVelo
Gene-Level Cosine Similarity
cellDancer SDEvelo scVelo(dyn) velocyto dynamo UniTVelo veloVI DeepVelo(GCN) scVelo(sto) STT biVI DeepVelo(VAE) VeloVAE LatentVelo
cellDancer SDEvelo UniTVelo velocyto biVI DeepVelo(GCN) scVelo(dyn) dynamo VeloVAE veloVI DeepVelo(VAE) LatentVelo scVelo(sto) STT
cellDancer SDEvelo VeloVAE biVI DeepVelo(GCN) UniTVelo velocyto scVelo(sto) dynamo DeepVelo(VAE) scVelo(dyn) veloVI LatentVelo

Extended Data Fig. 4 | Cosine similarity on simulation datasets

a, Gene-level cosine similarity between predicted high-dimensional RNA velocity and simulated truth on 42 dyngen datasets. b, Cell-level cosine similarity in PCA2 on the same 42 datasets. The original simulation-comparison layout is retained with current data: alternating grey rows, blue bubble matrices and Median/Mean bars. Dot area is normalized within each panel using the global minimum minus 0.01 and maximum; blue fill shows within-dataset rank (lighter is better). Median/Mean summarize raw dataset scores; their bar widths share a minimum-minus-0.01/max display scale within each panel. Numbers mark the top six ranks, with ties retained. Display normalization does not change the raw cosine values. c, Updated cell-level (upper row) and gene-level (lower row) cosine similarities in the original cellDancer mono-kinetic, transcriptional-boost and multi-forward-branching scenarios at noise 0.2. The original vertical boxplot design is retained; methods are ordered by median within each subplot. Boxes show median and interquartile range, whiskers extend to 1.5 IQR, and outliers are shown. These are the existing common-finite-gene re-evaluation scores provided in the accompanying source workbook, not values digitized from historical artwork. Each scenario has one simulated dataset with 2,000 source cells; the source workbook records shared gene counts and valid denominators. Cells and genes are within-dataset observations, not independent simulation replicates. Methods without valid results are omitted from the corresponding subplot. Saved preprocessing differs by method; alignment does not establish physical velocity units. TopicVelo values in a-b retain their historical-table provenance limitation and are not independently verified native-transition estimates. The full noise-series results are retained in the source-data summary. No model was rerun and the composite ranking is unchanged.